Department of Computer Science and Engineering
BRAC University
CSE 260: Digital Logic Design

Experiment # 3: Parity Bit Checker and Generator

Objective:

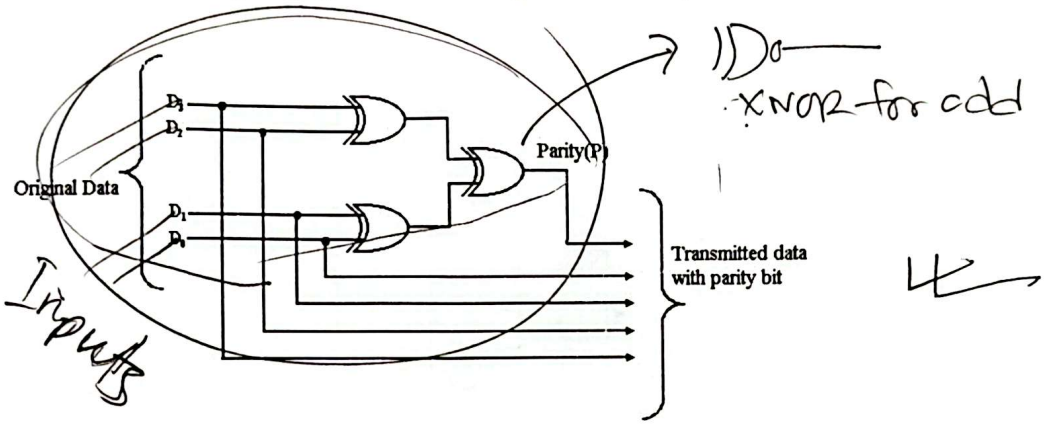
- To design and implement an even parity Generator and even parity checker using XOR gates. (IC-7486).

Required Components:

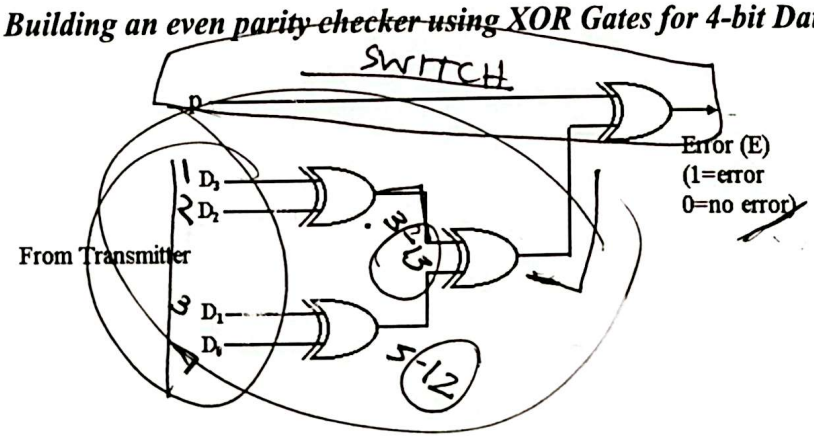
1. IC 7486 × 1

Diagrams:

Building an even parity generator using XOR Gates for 4-bit Data:



Building an even parity checker using XOR Gates for 4-bit Data:



Procedure:

- Construct the Circuits of both the parity bit generator and checker on the breadboard.
- Remember that each IC's pin 14 is connected to the "+5V" position of the DC Power Supply of AT-700, and pin 7 is connected to the "GND" position.
- Connect the inputs to the Data switches and outputs to any position of the LED Display.

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Result:

Complete both of the following truth tables.

Truth Table for Parity Bit Generator:

	Data				Parity
	D_3	D_2	D_1	D_0	
a.	1	0	0	1	0
b.	0	0	0	1	1
c.	1	1	1	1	0
d.	0	0	0	0	0

Truth Table for Parity Bit Checker:

	Data					Error
	Parity	D_3	D_2	D_1	D_0	
a.	1	1	0	0	1	1
b.	0	0	0	0	1	~~0~~1
c.	0	1	1	1	1	0
d.	1	0	0	0	0	1

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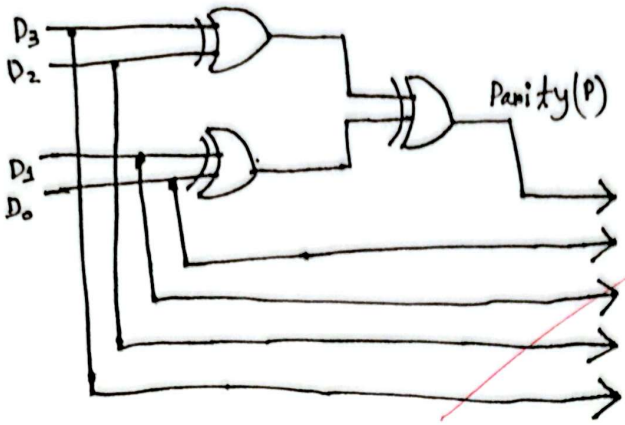

Figure: Even parity generator using XOR Gates for 4-bit Data

Truth table for parity bit generator

D3	D2	D1	D0	Parity
1	0	0	1	0
0	0	0	1	1
1	1	1	1	0
0	0	0	0	0

Truth table for parity bit checker

P	D3	D2	D1	D0	Error
1	1	0	0	1	~~0~~ 1
0	0	0	0	1	1
0	1	1	1	1	0
1	0	0	0	0	~~0~~ 1

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4-bit Data Even Parity

D_3	D_2	D_1	D_0	P
1	1	0	0	0
1	0	1	1	1

LSB → right
MSB → left

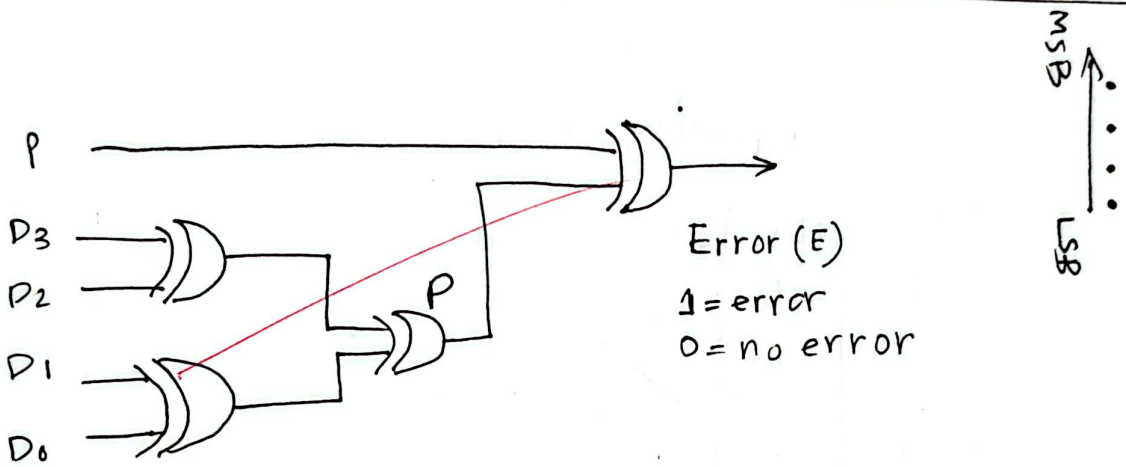

Fig: Parity checker using XOR for 4 bits

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